# Serving the full model. Meeting the latency gate.

## Qwen3-235B-A22B
## PD disaggregated inference

### VNUHCM - TripleU

Evidence snapshot | 12 October 2026

**BEST INTERNAL RESULT**

**1,912.30**

good output tokens/s
97.44% SLA fraction
N48 / 16 H200

Local measurements; organiser re-run and accuracy validation pending.

# Best measured stack, with explicit limits

### GOOD OUTPUT RATE

**1,912.30**

req/s goodput × 200
9.56150 good req/s

### REQUESTS WITHIN BOTH SLAs

**97.44%**

1,180 / 1,211 requests
0 errors; 0 cancellations

### VALIDATION STATUS

**Pending**

Full-set accuracy
Organiser official re-run

Evidence: job 22415 / confirm / N48; canonical AIPerf export.

# Throughput counts only when latency passes

### WORKLOAD

## 4,000 → 200

Synthetic Shakespeare inputs
Official FP8 checkpoint

### PER-REQUEST SLA

## 3,000 / 20

TTFT ≤ 3,000 ms
Mean ITL ≤ 20 ms

### POINT ACCEPTANCE

## ≥ 90%

Requests must pass both
Errors stay in denominator

Closed-loop load: a completed request is immediately replaced, sustaining N concurrent requests.

Task Rules + 2026-09-25 addendum, included under AI_Task/logs/rules/.

# One platform, one unchanged checkpoint

| | |
|---|---|
| **COMPUTE** | 2 × HGX H200 nodes / 16 GPUs total |
| **NETWORK** | InfiniBand NDR between the two nodes |
| **MODEL** | Qwen3-235B-A22B-Instruct-2507-FP8 |
| **SERVING** | SGLang 0.5.17 / router 0.3.2 / NIXL 1.3.2 |
| **CLIENT** | AIPerf 0.12.0; pinned official tokenizer |
| **PRECISION** | Published FP8 weights / BF16 KV (auto, verified in logs) |

Image metadata, SHA256 and model metadata: AI_Task/job_scripts/environment.json.

# Best layout: 2 prefill + 1 decode worker

## NODE 0 / 8 H200

**Prefill worker 0 · 4 GPUs · TP4 / EP1**

**Prefill worker 1 · 4 GPUs · TP4 / EP1**

**KV cache NIXL →**

## NODE 1 / 8 H200

**Decode worker 0**
**8 GPUs · TP8 / EP8**

AIPerf N48 → OpenAI-compatible gateway :8000 → Prefill → Decode → streamed tokens

P8-TP4_D8-TP8-EP8 / N48; hetero-tp-request-descriptors-v1; job 22415.

# Starting layouts establish our measured reference

## Reference layout 07

Prefill: 4 workers × TP2
Decode: 2 workers × TP4 / EP4
Node split: 8 + 8 GPUs
Long run: N32

## Previous best layout 09

Prefill: 2 workers × TP4
Decode: 2 workers × TP4 / EP4
Node split: 8 + 8 GPUs
Long run: N40

Trade-off: fewer prefill replicas, wider tensor parallelism; measure the balance under the SLA.

07 and 09 are organiser-provided starting layouts; the team extends the search and transport.

# A 17-layout search, with measured coverage

| GPU SPLIT P + D | PREFILL TP | DECODE TP | LAYOUTS |
|---|---|---|---|
| 8 + 8 | 2 / 4 / 8 | 2 / 4 / 8 | 9 |
| 4 + 12 | 2 / 4 | 2 / 4 | 4 |
| 12 + 4 | 2 / 4 | 2 / 4 | 4 |

## 14/17 originals measured

## + 3 measured EP2 variants

Each completed topology has N16, N32, N40 and N48 actual measurement exports.

AI_Task/logs/COVERAGE.md; pending and failed runs are not counted as completed measurements.

# Screen quickly; confirm at the full duration

## Deploy + initialize

All P/D paths
Separate setup prompts

## Explore

N16 / 32 / 40 / 48
10s + 20s + 30s

## Long confirmation

Best N48 / latest N32
60s + 120s + 30s

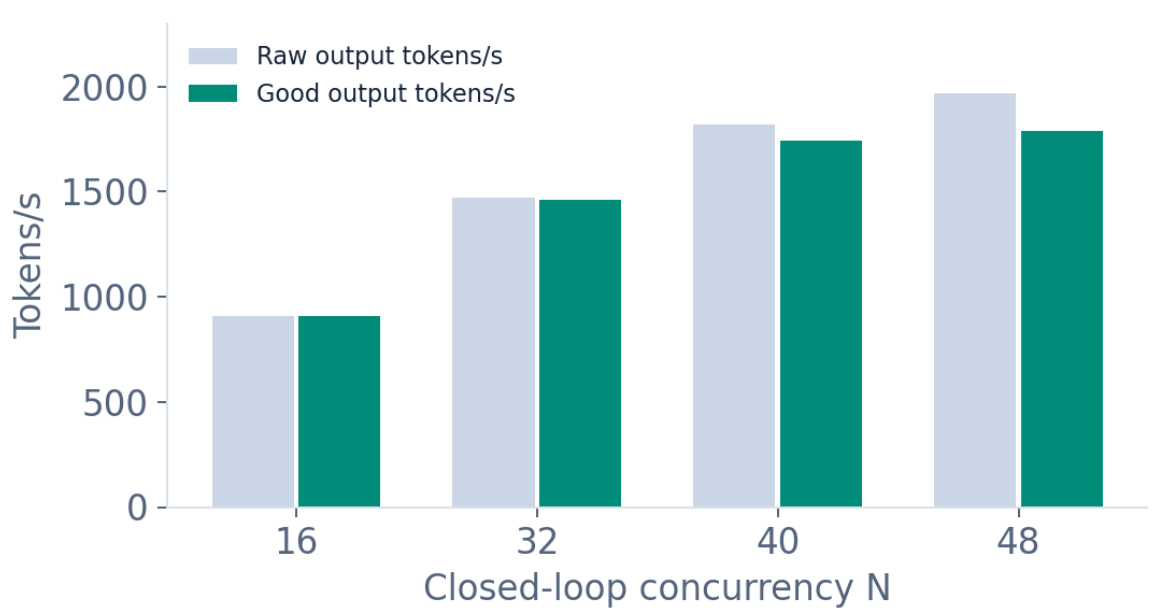
**Official timing**

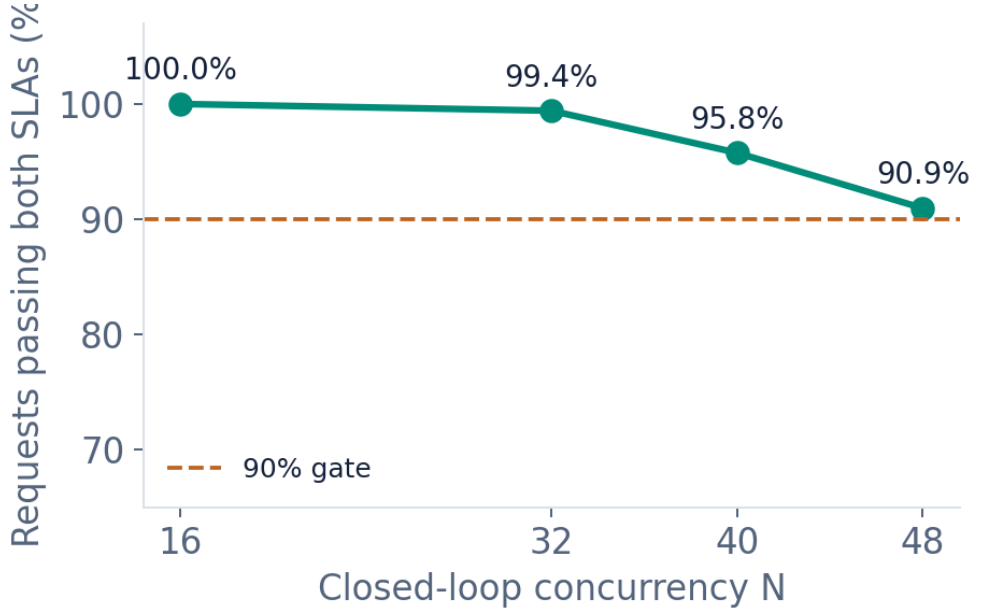



A short sweep and a long confirmation use different windows; their difference is not an optimisation gain.

Exploratory 10/20/30 vs long 60/120/30; original canonical exports retained.

# Winner sweeps select N48; the long run confirms it

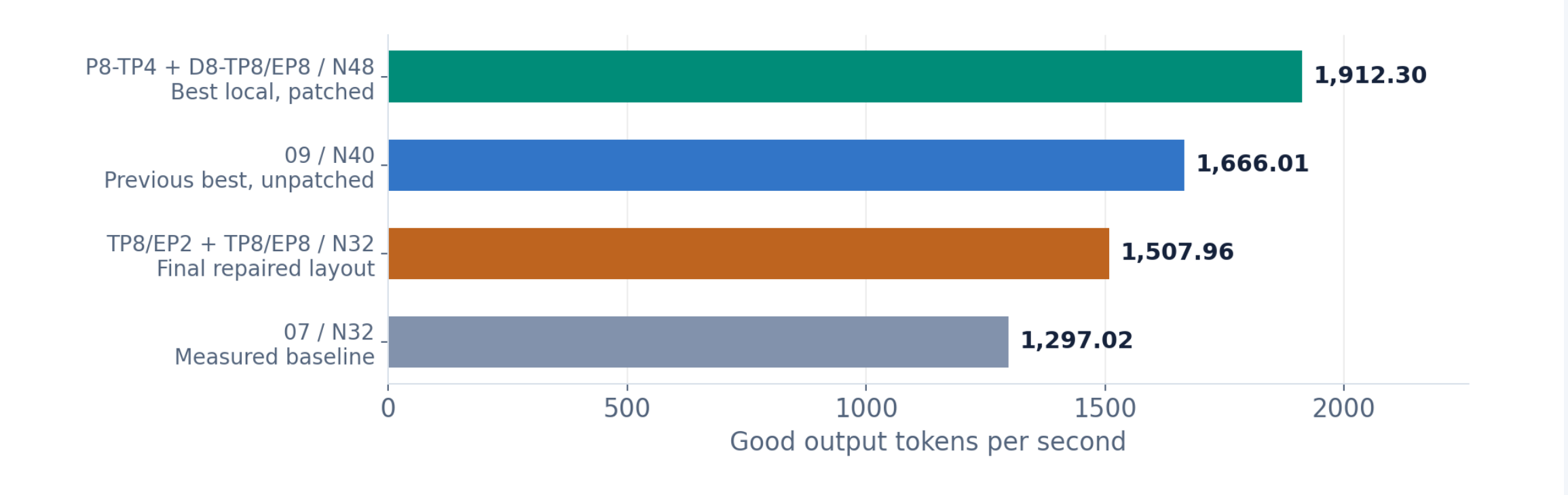



**N48 sweep: 1,788.65 good tokens/s; 90.95% SLA. Long N48: 1,912.30; 97.44% SLA.**

Job 22415 exploratory sweep: 10s warmup / 20s measurement / 30s grace.

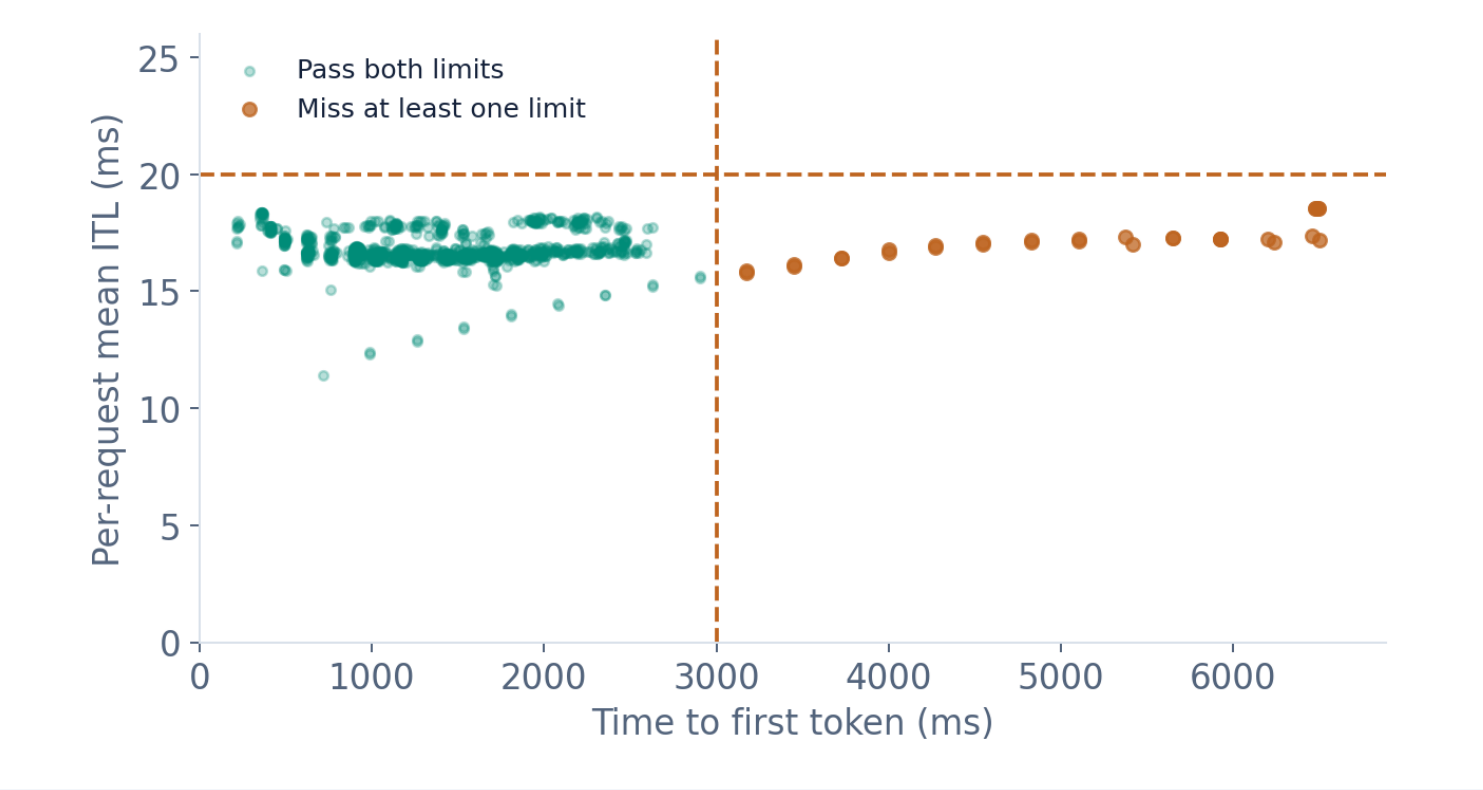
# Best long run: +14.78% over the previous best




Observed gain includes topology, concurrency and transport changes. Repeated matched A/B is still needed.

Jobs 22415, 22219, 22809 and 22197: long 60/120/30 runs on 16 H200.

# Check the gate per request, not from averages




**1,180 / 1,211**

requests pass both limits

Mean TTFT: 1,482.22 ms
Mean ITL: 16.79 ms
P95 TTFT: 2,456.56 ms
P95 ITL: 17.99 ms

Job 22415 / confirm N48: 1,211 profiling-phase records in the original JSONL.

# Failures remain evidence; actual exports count

| | | |
|---|---|---|
| **FP8 sharding** | TP8 / EP1 gives a 192-wide expert shard; FP8 block size is 128. | Use a distinct EP2 variant; keep the EP1 original incompatible. |
| **Cold P/D transfer** | 22778 timed out before measurement; patched 22809 completes. | Request-scoped descriptors; setup deadlines do not relax benchmark SLAs. |
| **Distributed ports** | A deployment hit EADDRINUSE before producing metrics. | Assign explicit distinct ports; preserve the failed logs and source. |

Immutable payloads and failed logs remain included; recovery registry distinguishes EP2 variants.

# EP2 makes TP8 compatible with FP8 blocks

## TP8 / EP1

128 experts / 1 expert group
1536 ÷ 8 = 192
192 is not divisible by 128

## TP8 / EP2

64 experts / group × 2 groups
1536 ÷ 4 = 384
384 = 3 × 128

All 3 EP2 replacements are measured. All experts and weights are retained; the originals stay labelled.

Checkpoint: MoE intermediate size 1536, 128 experts; image loader requires block_n = 128.

# NIXL: two request-scoped transport patches

## Full-pool descriptor plan

**≈ 403.8 million source entries / rank**

## Request-scoped plan

**188 entries: contiguous TP8→TP4 or TP8→TP8**

Byte ranges, head mapping and notifications retained.

**720 + 936**

CPU range cases
Both patches GPU-tested

**1,507.96**

latest equal-TP / N32
95.38% SLA; no errors

Heterogeneous TP: jobs 22294 / 22415. Equal TP8: job 22809. Original source and diffs included.

# The primary job deploys the winning stack

| | | |
|---|---|---|
| 01 | **Winning layout** | 2 prefill TP4 / EP1 + 1 decode TP8 / EP8; N48; all 16 H200. |
| 02 | **Enabled patch** | Read-only NIXL module mount; source hash checked by every worker. |
| 03 | **One Slurm job** | Deploy, initialize both P/D paths, run the locked official command once. |
| 04 | **Reconstruction** | Exact organiser images; published model; portable paths; offline cache. |

AI_Task/job_scripts/submit.sbatch; README.md; nixl.patch; measured job 22415.

# Correctness claims follow the available evidence

| | |
|---|---|
| Published FP8 weights and BF16 KV, unchanged | **Worker logs / source** |
| Plain PD on 16 GPUs; no speculative decoding | **Configured / local logs** |
| Best long run / N48 latency gate | **97.44% pass fraction** |
| Full-set accuracy ≥ 99% of reference | **Not yet tested** |
| Organiser reconstruction + official scored re-run | **Pending** |

Local latency evidence is available. Official acceptance and any Bonus validation remain pending.

Qwen addendum Q1/Q2; accuracy pass line: at least 99% of organiser reference score.

# Topology coverage complete; submission work remains

## Measured coverage

14 original + 3 EP2 layouts
All four concurrency sweeps
Winning long-run evidence
Both NIXL patches and logs

## Still outstanding

Full-set accuracy check
Packaged job GPU replay
HPC/OpenFOAM evidence
Official upload and re-run

Snapshot: 2026-10-12T11:17:26.130353+00:00. Job 22809 completed 11 October, 23:18 UTC.

# Current winner: TP4 prefill + TP8 / EP8 decode

**1,912.30**

good output tokens per second

97.44% request SLA fraction · N48 / 16 H200 · Official FP8 model

Accuracy and organiser re-verification remain pending.

Job 22415 / N48; hetero-tp-request-descriptors-v1; long 60/120/30 measurement.

# Appendix: the final repaired layout and selection

| | |
|---|---|
| **Final layout** | P8-TP8-EP2_D8-TP8-EP8; job 22809 completed. |
| **N32 long run** | 1,507.96 good tokens/s; 95.38% SLA; no errors/cancels. |
| **N40 / N48 short sweeps** | 89.50% / 86.06% SLA: both miss the 90% gate. |
| **Selection** | Job 22415 remains the best passing long measurement. |
| **Interpretation** | Measured coverage does not mean every layout passes SLA. |

Job 22809 / equal-tp8-request-descriptors-v1; sweeps 10/20/30 and confirm 60/120/30.